



SACS

Superelement

Version 24.00

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1 INTRODUCTION

The Superelement module is designed to be a multi-purpose module that will perform a variety of functions in regard to substructures or superelements. A substructure is defined to be a portion of structure which has been modeled and reduced down to a set of boundary joints. The total effect of a substructure on the rest of the structure can be represented in terms of the substructure reduced stiffness matrix and the reduced forces. Substructuring can be useful where the structure is too large for a single analysis, where a portion of the structure will not be changing for various configurations or where portions of the structure are repeated.

The substructure program module can be used for the following functions:

1. Creation of substructure
2. Expansion of retained model results
3. Changing load case naming
4. Adding in zero load cases
5. Changing boundary joint naming
6. Merging two substructures into one
7. Inputting stiffness directly into an analysis
8. Formatting the superelement: This function converts SACS superelement binary file into a formatted (ASCII) text file. This option may be used to *export* SACS superelement to other programs.
9. Unformatting the superelement: This function converts a formatted ASCII file into SACS binary superelement file. This function provides a convenient method to *import* superelement data into SACS.

2 ANALYSIS PROCEDURE

The analysis procedure depends on the function that the substructure is performing.

2.1 CREATING A SUPERELEMENT

The first function is in the creation of a substructure or superelement. In the SACS system, the substructure information is contained in a substructure file. This file basically contains 6X6 stiffness partitions and the 6Xn reduced load vectors where n is the number of load cases.

In order to create a superelement, the boundary joints degrees of freedom must be designated as retained degrees of freedom by using '222222' as the fixity. The Create Superelement option must be designated by either specifying 'C' in column 10 on the OPTIONS line in the substructure model or by selecting the appropriate option in the Executive.

Note: No superelement input file is required to create a superelement.

The substructure down is reduced down to these boundary joints. This procedure mathematically is as follows. The Reduced Stiffness Matrix $\mathbf{K}_{rr}$ is computed as

$$\mathbf{K}_{rr} = \mathbf{K}_{RR} - \mathbf{K}_{RF} \mathbf{K}_{FF}^{-1} \mathbf{K}_{FR}, \quad (1)$$

and Reduced Forces $\mathbf{f}_r$ are

$$\mathbf{f}_r = \mathbf{f}_R - \mathbf{K}_{RF} \mathbf{K}_{FF}^{-1} \mathbf{f}_F, \quad (2)$$

where R and F denote retained and free degrees of freedom, respectively.

The superelement file contains the reduced stiffness matrix, reduced force vectors and other information required for the subsequent expansion.

After condensing or reducing the substructure, the retained model should be solved using the substructure file as input along with the retained model file. The model file must contain the boundary joints that were designated in the substructure file. However it should not contain the constrained (reduced) joints of the substructure.

Note: The boundary joints in the retained model do not need any special boundary condition specifications.

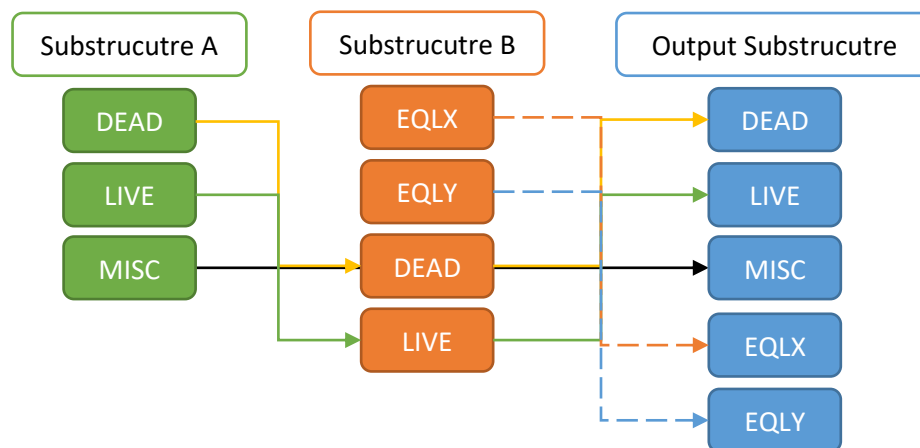
The load conditions for the retained model must be the same as for the substructure. The solution file will contain the deflections of all the joints in the retained model including the boundary joints of the substructure.

After solving, the results may be expanded where the deflections of the boundary joints are extracted from the retained model results. These deflections can then be used to obtain all the deflections of the substructure.

2.2 MERGING SUPERELEMENTS

In addition to creating and expanding substructures, the Superelement module can be used as a substructure utility program to merge substructures. The merging of two substructures results in a substructure containing the stiffness and forces from both substructures. The process of merging can be continued indefinitely so that any number of substructures can be merged. It should also be noted that substructures can be used to create other substructures so that a substructure can be a part of another substructure. This procedure can be repeated to accomplish any level of substructuring.

The superelement program merges the load cases based on their names by default. If the load case name is the same in both substructures, the Superelement program combines two loads into a single load case in the output substructure. If a given load case name is not present in the other substructure, the program simply transfers the load case to the output superelement. The following chart illustrates the automatic process of merging load cases.



Note: No superelement input file is required when automatically merging substructures.

2.2.1 Manually Merging Loading

The MRGLOAD input line can be used to *manually* merge load cases from two substructures by specifying their labels on this input line. This line can be repeated as many times as necessary to merge all required load cases. The MRGLOAD input line merges load cases as follows:

- Merge two load cases with different names: Enter the load case name pair on the MRGLOAD line and the program combines two load cases. The output load case name will be the same as the load case name in the first superelement file.
- Transfer load cases without merging: Enter the name of the load case being transferred from the first superelement file on the MRGLOAD line, while leaving the name for the second superelement *blank*, and vice versa.

Note: The load cases in the output substructure will be the same as the load name in the first substructure if it is not blank. otherwise, the output load case name is the same as the second substructure.

The following sample MRGLOAD line performs the following load merge:

1. Merge load case 'FX' from the first substructure with the 'LX' load case from the second substructure and transfer it as 'FX' to the output substructure.
2. Transfer load case 'FY' with the same name to the output substructure.
3. Transfer load case 'LY' with the same name to the output substructure.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	MRGLOAD	FX	LX	FY	LY			

Note: A superelement input file is required when merging load cases using the MRGLOAD input line.

Note: The program only includes the load cases entered in the line. Other load cases are not transferred to the new substructure. therefore, all load cases that are to be included in the new substructure must be specified.

2.3 Modifying Superelement

The Superelement program can modify a superelement file by renaming the joints and the load cases using CHGJ and CHGL input lines, respectively. The modifying superelement function is implemented by entering MOD on columns 8-10 of the SUBOPT input line.

2.3.1 Modifying Joint Labels

Joint labels can be renamed by entering the original name and new name of the joint on the CHGJ input line. A total of 9 joints can be entered on a single CHGJ line and it can be repeated as many times as necessary to modify the required joint labels. For the joints that are not listed on this line, the Superelement program uses their original names in the input superelement.

Note: If the program does not find a joint in the original superelement, it silently ignores the input.

The following example illustrates how the CHGJ line is used to rename joints 0001, 0002, 0003, to P001m P002, and P003, respectively.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	CHGJ	0001	P001	0002	P002	0003	P003	

2.3.2 Modify Load Cases

The Superelement Program can modify a superelement by renaming, removing, or adding empty (zero) load cases, using the CHGL input line.

- **Renaming a load case:** Enter the original name of the load case in the input superelement and its new name on the CHGL input line, and the program renames the load case.
- **Removing a load case:** Do not list the load case name on the CHGL input line. The program only transfers the load cases listed on the CHGL line to the modified superelement and ignores the rest.

Note: All load cases that are to be included in the new substructure must be specified even though its load case name is the same.

- **Adding an empty (zero) load case:** Leave the original name of the load case blank while entering a new name for the load case on the CHGL line. The program automatically adds a zero (empty) load case to the output substructure.

A total of 9 load cases can be entered on a single CHGL line and it can be repeated as many times as necessary to modify the required load cases.

The following example shows how to rename load cases 0001, and 0002 to DEAD and LIVE respectively, and add an empty load case EQLX.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
2	CHGL	0001DEAD	0002LIVE	EQLX				

2.4 Inputting Stiffness Directly

The Superelement program may also be used to input stiffness matrices directly into the SACS solution. The Superelement input file must contain the stiffness terms to be included.

Note: When inputting stiffness directly, the header must be the INPUT line.

2.5 Formatting Superelement

The Superelement program is able to convert superelement data (stiffness matrix and load vectors) from a SACS binary file to a formatted (ASCII) text file – which can be used to export SACS superelement to other analysis programs. This option is available by selecting ‘FRM’ in columns #8-10 of SUBOPT input line.

This functionality of the Superelement program provides two new options to format the output text file: 1) rotating the coordinate system and 2) re-ordering degrees of freedom (DOF) in output stiffness matrix and load vectors.

The coordinate system can be rotated by setting the vertical axis direction on SUBOPT input line. The user may input the vertical coordinate direction in column # 17-18 of SUBOPT line. The user may input following values: (+X, -X, +Y, -Y, +Z, -Z) or leave it blank for the default value (i.e. +Z). The unit of output data can be also set on SUBOPT line in columns # 12-13.

The order of DOF in output superelement can be modified using **DOFORD** input line. The user may input new DOF order in column # 8-13 of DOFORD input line. The default order of DOF in SACS is ‘RX RY RZ DX DY DZ’ – where R denotes rotation about an axis and D is displacement along an axis. The user may enter a six-digit number to change the order of DOF. For example, the input line “DOFORD 456123” re-orders the output degrees of freedom as ‘DX DY DZ RX RY RZ’.

Note: DOFORD input line is optional and if it is not present in the input file, the Superelement program returns superelement matrix and load vectors with default SACS order – i.e. ‘RX RY RZ DX DY DZ’

The output ASCII superelement file (with default name *subsef.**) has two main partitions: 1) the Stiffness Matrix and 2) the Load Vectors. The stiffness section starts with comment lines (starting with *) to provide information regarding the size of the stiffness matrix. Next, the size of the stiffness matrix is printed and the *full* stiffness matrix is immediately written.

Similar to the stiffness section, the Load Vector section starts with additional information about the load vectors, and then the number of load conditions are printed followed by the load vectors. Each row of load vector data is corresponding to a given load condition at all degrees of freedom.

Note: The load section is only printed, if at least one load condition is defined in superelement.

Following figure illustrated a sample of formatted (ASCII) SACS superelement.

```
*
* SACS Formatted Superelement
* Note: For the Order of Degrees of Freedom check the Listing File.
* Note: For output units check the Listing File.
* Size of Superelement Full Stiffness Matrix - 6 x Number of Joints:
12
0.79259135E+07 0.00000000E+00 -0.33780434E+07 0.00000000E+00 -0.22165655E+05 0.00000000E+00 -0.79259151E+07 0.00000000E+00 -0.33780434E+07 0.00000000E+00 0.22165651E+05 0.00000000E+00
0.00000000E+00 0.22181759E+08 0.00000000E+00 0.69443859E+05 0.00000000E+00 -0.29750832E+05 0.00000000E+00 -0.13113711E+08 0.00000000E+00 -0.69443857E+05 0.00000000E+00 0.29750832E+05
-0.33780434E+07 0.00000000E+00 0.53811840E+07 0.00000000E+00 0.15912695E+05 0.00000000E+00 0.33780434E+07 0.00000000E+00 -0.53099701E+06 0.00000000E+00 -0.15912695E+05 0.00000000E+00
0.00000000E+00 0.69443859E+05 0.00000000E+00 0.34175143E+03 0.00000000E+00 0.33921842E+11 0.00000000E+00 -0.69443857E+05 0.00000000E+00 -0.34175144E+03 0.00000000E+00 -0.34058080E+11
-0.22165655E+05 0.00000000E+00 0.15912695E+05 0.00000000E+00 0.10441410E+03 0.00000000E+00 0.22165651E+05 0.00000000E+00 0.15912695E+05 0.00000000E+00 -0.10441410E+03 0.00000000E+00
0.00000000E+00 -0.29750832E+05 0.00000000E+00 0.33921842E+11 0.00000000E+00 0.19521025E+03 0.00000000E+00 -0.29750832E+05 0.00000000E+00 0.36543949E+11 0.00000000E+00 -0.19521553E+03
-0.79259151E+07 0.00000000E+00 0.33780434E+07 0.00000000E+00 0.22165651E+05 0.00000000E+00 0.79259135E+07 0.00000000E+00 0.33780434E+07 0.00000000E+00 -0.22165655E+05 0.00000000E+00
0.00000000E+00 -0.13113711E+08 0.00000000E+00 -0.69443857E+05 0.00000000E+00 -0.29750832E+05 0.00000000E+00 0.22181759E+08 0.00000000E+00 0.69443859E+05 0.00000000E+00 0.29750832E+05
-0.33780434E+07 0.00000000E+00 -0.53099701E+06 0.00000000E+00 0.15912695E+05 0.00000000E+00 0.33780434E+07 0.00000000E+00 0.53811840E+07 0.00000000E+00 -0.15912695E+05 0.00000000E+00
0.00000000E+00 -0.69443857E+05 0.00000000E+00 -0.34175144E+03 0.00000000E+00 0.36543949E+11 0.00000000E+00 0.69443859E+05 0.00000000E+00 0.34175143E+03 0.00000000E+00 -0.35679893E+11
0.22165651E+05 0.00000000E+00 -0.15912695E+05 0.00000000E+00 -0.10441410E+03 0.00000000E+00 -0.22165655E+05 0.00000000E+00 -0.15912695E+05 0.00000000E+00 0.10441410E+03 0.00000000E+00
0.00000000E+00 0.29750832E+05 0.00000000E+00 -0.34058080E+11 0.00000000E+00 -0.19521553E+03 0.00000000E+00 0.29750832E+05 0.00000000E+00 -0.35679893E+11 0.00000000E+00 0.19521025E+03
* Load Conditions
* Note: For Load Condition Names check the Listing File.
* Note: For output units check the Listing File.
* Number of Load Conditions:
* Load Vectors - Rows a load condition, Column: Degrees of Freedom
* Size: Rows - Number of Load Conditions, Columns: 6 x Number of Joints:
2
0.00000000E+00 -0.39704857E+06 0.00000000E+00 -0.22737391E+04 0.00000000E+00 -0.19383151E+04 0.00000000E+00 -0.39470283E+06 0.00000000E+00 -0.22421902E+04 0.00000000E+00 0.19383151E+04
0.36966293E+06 0.00000000E+00 -0.11475251E+06 0.00000000E+00 -0.75296959E+03 0.00000000E+00 0.10128876E+07 0.00000000E+00 -0.11475251E+06 0.00000000E+00 -0.37829597E+04 0.00000000E+00
```

In addition to formatted superelement file, the Superelement program provides additional information about joint IDs, order of DOFs, output units, and name of load conditions in the analysis Listing File. A sample of report in the Listing File is shown below.

```
===== Converting SACS Superelement to Formatted Text File =====

----- Superelement Unit Information -----

Force Unit is "KG", Moment Unit is "KG x cm", Displacement Unit is "cm".

--Joints and Degrees of Freedom Information of the Superelement--

      DOF #      Joint Name      DOF ID
      -----      -
      1          0000          RX
      2          0000          RY
      3          0000          RZ
      4          0000          DX
      5          0000          DY
      6          0000          DZ
      7          0003          RX
      8          0003          RY
      9          0003          RZ
     10          0003          DX
     11          0003          DY
     12          0003          DZ

----- Load Conditions Information of the Superelement -----

      Load #      Load Name
      -----      -
      1          L1
      2          L2
```

2.6 Unformatting Superelement

The Superelement program Unformatting is a functionality to convert and import formatted superelement file into the SACS. This function becomes available by selecting 'UNF' in column #8-10 of SUBOPT input line. The data corresponding unformatting functionality is entered into two steps: 1) Superelement program input file (with default name of *subinp.**), and 2) Formatted ASCII Superelement Data file (with default name of *subsef.**).

Step 1: The Superelement program input file contains following inputs: Superelement joint IDs (mandatory) and Superelement load conditions (optional). The joint IDs are entered using following two input lines:

INTJNT (interface joint(s) header line): This line is used to input number of interface joints in the superelement. This line should be followed by JOINT input line.

- ✓ Column # 8-10: number of interface joints which the program computes the stiffness matrix.

JOINT: This line is used to input joint names of the interface joints. The user must input one JOINT line for each interface joint.

The load conditions may be similarly entered using below input lines:

LOAD (load condition(s) header line): This line is used to input number of load conditions in the superelement. This line should be followed by LOADCN input line.

- ✓ Column # 8-10: number of load conditions which the program computes the load vectors.

LOADCN: This line is used to input load condition names. The user must input one LOADCN line for each load condition.

Similar to Formatting function discussed in previous section, **DOFORD** input can be used to re-order digress of freedom of imported superelement – if the order is different from SACS default DOF order – i.e. 'RX RY RZ DX DY DZ'. For example, if the imported superelement stiffness has 'DX DY DZ RX RY RZ' as DOF order in a given joint, input line "DOFORD 456123" convert order of DOF to "RX RY RZ DX DY DZ".

Also, if the coordinate system of imported superelement is different with the default SACS coordinate system (i.e. +Z is vertical axis in SACS program), the vertical direction can be given in column #17-18 of SUBOPT input line. Additionally, the unit of imported superelement can be set on SUBOPT line in columns # 12-13.

A sample Superelement program input file with Unformatting function is presented below.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
3	SUBOPT	UNF	ME					
4	DOFORD	456123						
5	INTJNT	2						
6	JOINT	0001						
7	JOINT	0003						
8	LOAD	2						
9	LOADCN	DEAD						

```
10 LOADCN LIVE
11 END
```

Step 2: The second step involves entering the stiffness matrix and load vectors corresponding to the imported superelement. The input ASCII file should have a similar format discussed in previous section for *Formatting* function. The file has two sections: the first section is full stiffness matrix and the second section is load vectors (if any). Each section may has own comments lines.

The stiffness section may start with optional comment lines (these lines starts with *) followed by size of the full stiffness matrix. Then, the entire superelement stiffness matrix is entered. Similarly, the load condition may start with comment lines followed by number of load conditions. The load vector associated with each load condition is then entered immediately after number of load conditions. A sample of formatted ASCII input for importing superelement is given below.

```
* A sample for formatted superelement file
* Size of Superelement Full Stiffness Matrix - 6 x Number of Joints:
6
0.79259135E+07 0.00000000E+00 -0.33780434E+07 0.00000000E+00 -0.22165655E+05 0.00000000E+00
0.00000000E+00 0.22181759E+08 0.00000000E+00 0.69443859E+05 0.00000000E+00 -0.29750832E+05
-0.33780434E+07 0.00000000E+00 0.53811840E+07 0.00000000E+00 0.15912695E+05 0.00000000E+00
0.00000000E+00 0.69443859E+05 0.00000000E+00 0.34175143E+03 0.00000000E+00 0.33925862E-11
-0.22165655E+05 0.00000000E+00 0.15912695E+05 0.00000000E+00 0.10441410E+03 0.00000000E+00
0.00000000E+00 -0.29750832E+05 0.00000000E+00 0.33925862E-11 0.00000000E+00 0.19521025E+03
* Number of Load Conditions:
2
0.00000000E+00 -0.39704957E+06 0.00000000E+00 -0.22737391E+04 0.00000000E+00 -0.19383151E+04
0.36966293E+06 0.00000000E+00 -0.11475251E+06 0.00000000E+00 -0.75296959E+03 0.00000000E+00
```

Note: There is no comment line allowed between input line of the stiffness matrix size and matrix elements. There is no comments allowed between number of load conditions and load vectors either.

Note: Similar to Formatting, the listing file will provide additional information regarding order of degrees of freedom, units and load condition names of imported superelement data.

3 COMMENTARY

3.1 SUBSTRUCTURE CREATION

The creation of a substructure is basically the process of reducing the internal degrees of freedom so that the independent degrees of freedom correspond to the boundary joints. If the subscripts i and b represent the internal and boundary degrees of freedom respectively, then

$$\mathbf{f}_i = \mathbf{K}_{ii}\mathbf{d}_i + \mathbf{K}_{ib}\mathbf{d}_b, \quad (3)$$

and

$$\mathbf{f}_b = \mathbf{K}_{bi}\mathbf{d}_i + \mathbf{K}_{bb}\mathbf{d}_b. \quad (4)$$

Solving for interior degrees of freedom, we get:

$$\mathbf{C}_{ii} = \mathbf{K}_{ii}^{-1}, \quad (5)$$

$$\mathbf{d}_i = \mathbf{C}_{ii}(\mathbf{f}_i - \mathbf{K}_{ib}\mathbf{d}_b), \quad (6)$$

and

$$\mathbf{f}_b - \mathbf{K}_{bi}\mathbf{C}_{ii}\mathbf{f}_i = (\mathbf{K}_{bb} - \mathbf{K}_{bi}\mathbf{C}_{ii}\mathbf{K}_{ib})\mathbf{d}_b. \quad (7)$$

Therefore, the reduced force is:

$$\bar{\mathbf{f}} = \mathbf{f}_b - \mathbf{K}_{bi}\mathbf{C}_{ii}\mathbf{f}_i, \quad (8)$$

and the reduced stiffness is:

$$\bar{\mathbf{K}} = \mathbf{K}_{bb} - \mathbf{K}_{bi}\mathbf{C}_{ii}\mathbf{K}_{ib}. \quad (9)$$

Above $\mathbf{f}$ are force vectors, $\mathbf{K}$ are stiffness matrices, and $\mathbf{d}$ are deflection vectors.

The resulting reduced stiffness and force matrices may be used to represent the entire substructure even though only a subset of the total number of degrees of freedom are present. It may be considered as elemental matrix with many degrees of freedom hence the term 'superelement'.

3.2 SUBSTRUCTURE EXPANSION

The expansion of a substructure uses the relations as did the reduction phase. Mainly

$$\mathbf{d}_i = \mathbf{C}_{ii}(\mathbf{f}_i - \mathbf{K}_{ib}\mathbf{d}_b), \quad (10)$$

Since the boundary deflections ($\mathbf{d}_b$) can be obtained from the retained model results, the above equation allows the calculation of the internal degrees of freedom deflections which can then be used to calculate reactions, internal loads, etc.

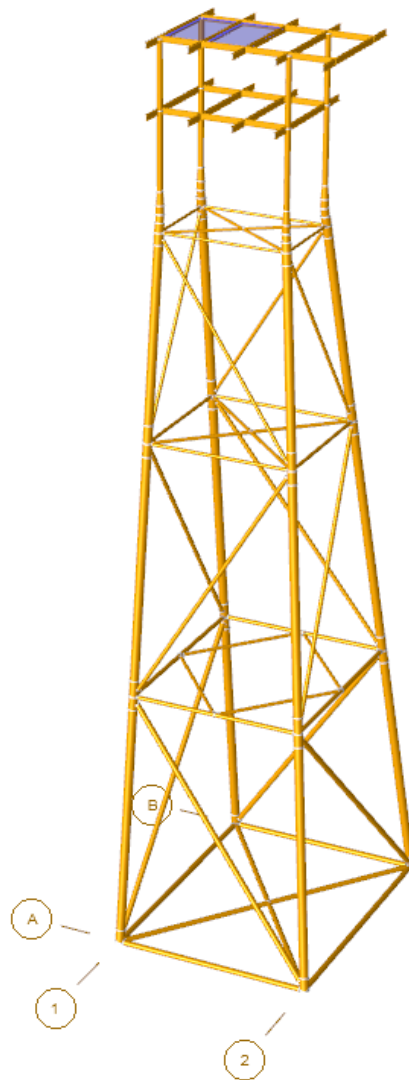
4 SAMPLE PROBLEMS

Three sample problems are illustrated for the Superelement program. Each of the problems use the structure shown below.

In Sample Problem 1, the structure is separated into two parts, the deck structure and the jacket structure.

Sample Problem 2 is an analysis of the jacket structure and expansion of the deck superelement for post-processing.

Sample Problem 3 illustrates direct stiffness input. In this example, the 6X6 stiffness partitions for joints 1, 2, 3, 4, 101, 102, 103, and 104 including the off-diagonal partitions are specified.



4.1 SUPERELEMENT CREATION

For this sample problem, the structure is to be separated into two parts, the deck structure and the jacket structure as shown below. The deck is the substructure and the boundary joints are 601, 603, 605, and 607. This is indicated in the deck model data by the boundary conditions of '222222' for these joints.

Note: These joints are also in the jacket model with no special boundary conditions.

The retained model (i.e. jacket model) is then solved using the deck superelement.



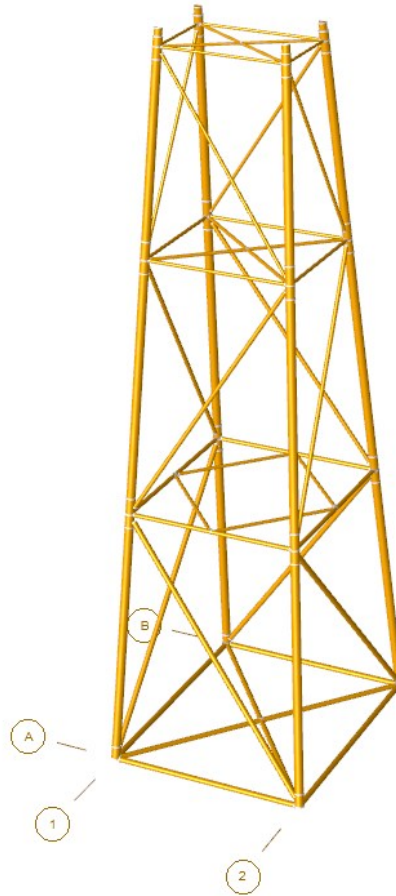
A portion of the deck model is shown below followed by an explanation of the input.

```
12345678901234567890123456789012345678901234567890123456789012345678901234567890
1  SAMPLE 13 ENGLISH UNITS MODEL SUPER ELEMENT (DECK)
2  *USE SEPERATE SEASTATE INPUT FILE
3  *CMB IS INPUTTED IN SEASTATE OPTION LINE
4  *"C" IN INPUTTED IN COLUMN 10 TO CREATE SUPER ELEMENT
5  OPTIONS C EN SDUC 2 1 PTPT PT
6  SECT
7  SECT CONE CON 36.0000.75026.000
8  GRUP
9  GRUP LG6 36.000 0.750 29.0011.0036.00 1 1.001.00 0.500N490.003.25
10 GRUP LG6 CONE 29.0011.6036.00 1 1.001.00 0.500N490.004.95
11 GRUP LG6 26.000 0.750 29.0011.6036.00 1 1.001.00 0.500N490.00
12 GRUP LG7 26.000 0.750 29.0011.6036.00 1 1.001.00 0.500N490.00
13 GRUP W01 W24X162 29.0111.2035.97 1 1.001.00 0.500 489.99
14 GRUP W02 W24X131 29.0111.2035.97 1 1.001.00 0.500 489.99
15 MEMBER
16 MEMBER 601 701 LG6
17 ***** Additional Member Data *****
18 PGRUP
19 PGRUP P01 0.3750I29.000 0.25036.000 490.0000
20 PLATE
21 PLATE AAAC 801 834 805 837 P01 0
22 PLATE AAAD 834 835 837 838 P01 0
```

23	JOINT								
24	JOINT 601	-24.	-16.	13.				222222	
25	JOINT 603	24.	-16.	13.				222222	
26	JOINT 605	-24.	16.	13.				222222	
27	JOINT 607	24.	16.	13.				222222	
28	JOINT 701	-24.	-16.	50.					
29	JOINT 703	24.	-16.	50.					
30	JOINT 705	-24.	16.	50.					
31	JOINT 707	24.	16.	50.					
32	JOINT 709	-24.	-26.	50.			-2.964		
33	JOINT 710	-8.	-26.	50.	-2.424		-2.964		
34	JOINT 711	8.	-26.	50.	2.424		-2.964		
35	JOINT 712	24.	-26.	50.			-2.964		
36	JOINT 714	-8.	-16.	50.	-2.424				
37	JOINT 715	8.	-16.	50.	2.424				
38	JOINT 717	-8.	16.	50.	-2.424				
39	JOINT 718	8.	16.	50.	2.424				
40	JOINT 720	-24.	26.	50.			2.964		
41	JOINT 721	-8.	26.	50.	-2.424		2.964		
42	JOINT 722	8.	26.	50.	2.424		2.964		
43	JOINT 723	24.	26.	50.			2.964		
44	JOINT 801	-24.	-16.	75.					
45	JOINT 803	24.	-16.	75.					
46	JOINT 805	-24.	16.	75.					
47	JOINT 807	24.	16.	75.					
48	JOINT 829	-24.	-26.	75.			-2.964		
49	JOINT 830	-8.	-26.	75.	-2.424		-2.964		
50	JOINT 831	8.	-26.	75.	2.424		-2.964		
51	JOINT 832	24.	-26.	75.			-2.964		
52	JOINT 833	41.	-26.	75.	0.132		-2.964		
53	JOINT 834	-8.	-16.	75.	-2.424				
54	JOINT 835	8.	-16.	75.	2.424				
55	JOINT 836	41.	-16.	75.	0.132				
56	JOINT 837	-8.	16.	75.	-2.424				
57	JOINT 838	8.	16.	75.	2.424				
58	JOINT 839	41.	16.	75.	0.132				
59	JOINT 840	-24.	26.	75.			2.964		
60	JOINT 841	-8.	26.	75.	-2.424		2.964		
61	JOINT 842	8.	26.	75.	2.424		2.964		
62	JOINT 843	24.	26.	75.			2.964		
63	JOINT 844	41.	26.	75.	0.132		2.964		
64	***** Additional Load Data *****								
65	END								

Line 5. On the OPTIONS input line, 'C' in column 10 indicates that a substructure is to be created.
 Lines 24-27. The boundary joints are 601, 603, 605, and 607 are indicated by the boundary conditions of '222222'.

After the superelement is created, the retained model or jacket model is solved. The retained model is shown in the figure below followed by a portion of the retained model input and an explanation of important data.



	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	SAMPLE 13 ENGLISH UNITS MODELSUPER ELEMENT (JACKET)							
3	* USE SEPERATE SEASTATE INPUT FILE							
4	* CMB IS INPUTTED IN SEASTATE OPTION LINE							
5	* "I" IS INPUTTED IN COLUMN 9 TO INDICATE NEED INPUT SUPER ELEMENT							
6	OPTIONS I	EN	SDUC	2 1	PTPT	PT		
7	GRUP							
8	GRUP LG1	42.000	1.375	29.0011.6050.00	1	1.001.00	0.500N490.005.00	
9	GRUP LG1	41.250	1.000	29.0011.6036.00	1	1.001.00	0.500N490.00	
10	GRUP LG1	42.000	1.375	29.0011.6050.00	1	1.001.00	0.500N490.005.00	
11	GRUP LG2	42.000	1.375	29.0011.6050.00	1	1.001.00	0.500N490.006.15	
12	GRUP LG2	41.250	1.000	29.0011.6036.00	1	1.001.00	0.500N490.00	
13	GRUP LG2	42.000	1.375	29.0011.6050.00	1	1.001.00	0.500N490.004.90	
14	GRUP LG3	42.000	1.375	29.0011.6050.00	1	1.001.00	0.500N490.006.75	
15	GRUP LG3	41.250	1.000	29.0011.6036.00	1	1.001.00	0.500N490.00	
16	GRUP LG3	42.000	1.375	29.0011.6050.00	1	1.001.00	0.500N490.004.35	
17	GRUP LG4	42.000	1.375	29.0011.6050.00	1	1.001.00	0.500N490.00	
18	GRUP LG5	36.000	1.000	29.0011.6050.00	1	1.001.00	0.500N490.00	
19	GRUP PL1	36.000	1.000	29.0011.6036.00	1	1.001.00	0.500N490.00	
20	GRUP PL2	36.000	1.000	29.0011.6036.00	1	1.001.00	0.500N490.00	
21	GRUP PL3	36.000	1.000	29.0011.6036.00	1	1.001.00	0.500N490.00	
22	GRUP PL4	36.000	1.000	29.0011.6036.00	1	1.001.00	0.500N490.00	
23	GRUP T01	16.000	0.625	29.0111.2035.00	1	1.001.00	0.500N490.00	
24	GRUP T02	20.000	0.750	29.0011.6035.00	1	1.001.00	0.500N490.00	
25	GRUP T03	12.750	0.500	29.0111.6035.00	1	1.001.00	0.500N490.00	
26	GRUP T04	24.000	0.750	29.0011.6036.00	1	1.001.00	0.500N490.00	
27	GRUP T05	26.000	1.000	29.0011.6036.00	1	1.001.00	0.500N490.00	
28	GRUP W.B	36.433	1.000	29.0111.2035.97	1	1.001.00	0.500 490.00	
29	MEMBER							

```

29 MEMBER 101 201 LG1
30 ***** Additional Member Data *****
31 JOINT
32 JOINT 101 -24. -50. -261. -3.000
33 JOINT 102 -24. -50. -261. -3.000 PILEHD
34 JOINT 103 51. -50. -261. 4.800 -3.000
35 JOINT 104 51. -50. -261. 4.800 -3.000 PILEHD
36 JOINT 105 -24. 50. -261. 3.000
37 JOINT 106 -24. 50. -261. 3.000 PILEHD
38 JOINT 107 51. 50. -261. 4.800 3.000
39 JOINT 108 51. 50. -261. 4.800 3.000 PILEHD
40 JOINT 109 13. 0. -261. 8.400
41 JOINT 201 -24. -38. -164. -1.500
42 JOINT 202 -24. -38. -164. -1.500
43 JOINT 203 41. -38. -164. 8.400 -1.500
44 JOINT 204 41. -38. -164. 8.400 -1.500
45 JOINT 205 -24. 38. -164. 1.500
46 JOINT 206 -24. 38. -164. 1.500
47 JOINT 207 41. 38. -164. 8.400 1.500
48 JOINT 208 41. 38. -164. 8.400 1.500
49 JOINT 209 -24. 0. -164.
50 JOINT 210 8. 38. -164. 10.296 1.500
51 JOINT 211 41. 0. -164. 8.400
52 JOINT 212 8. -38. -164. 10.296 -1.500
53 JOINT 301 -24. -26. -69. -3.000
54 JOINT 302 -24. -26. -69. -3.000
55 JOINT 303 32. -26. -69. 2.400 -3.000
56 JOINT 304 32. -26. -69. 2.400 -3.000
57 JOINT 305 -24. 26. -69. 3.000
58 JOINT 306 -24. 26. -69. 3.000
59 JOINT 307 32. 26. -69. 2.400 3.000
60 JOINT 308 32. 26. -69. 2.400 3.000
61 JOINT 309 4. 0. -69. 1.200
62 JOINT 401 -24. -16. 6. -9.756 6.000
63 JOINT 402 -24. -16. 6. -9.756 6.000
64 JOINT 403 24. -16. 6. 7.800 -9.756 6.000
65 JOINT 404 24. -16. 6. 7.800 -9.756 6.000
66 JOINT 405 -24. 16. 6. 9.756 6.000
67 JOINT 406 -24. 16. 6. 9.756 6.000
68 JOINT 407 24. 16. 6. 7.800 9.756 6.000
69 JOINT 408 24. 16. 6. 7.800 9.756 6.000
70 JOINT 409 0. 0. 6. 3.900 6.000
71 JOINT 501 -24. -16. 10. -4.500
72 JOINT 503 24. -16. 10. 3.600 -4.500
73 JOINT 505 -24. 16. 10. 4.500
74 JOINT 507 24. 16. 10. 3.600 4.500
75 JOINT 601 -24. -16. 13.
76 JOINT 603 24. -16. 13.
77 JOINT 605 -24. 16. 13.
78 JOINT 607 24. 16. 13.
79 ***** Additional Load Data *****
80 END

```

Line 5. An 'I' is specified in column 9 on the OPTIONS line indicating a substructure is to be imported.

Lines 75-78. The boundary joints appear in the master model without any boundary conditions

4.2 SUPERELEMENT EXPANSION

A static analysis of the jacket is performed with the deck included as a superelement. An additional seastate input file is used to define the wave loading on the structure and load combinations. The MISC, EQPT, AREA and LIVE load combinations were defined in the deck input file and with placeholder load conditions defined in the jacket input file. The following is a copy of the seastate input file:

```

1      12345678901234567890123456789012345678901234567890123456789012345678901234567890
2      LDOPT      NF+Z      64.30  490.00  -261.0  261.00GLOBEN      CMB  NPNP  K
3      * SELECTS STATIC ANALYSIS LOAD CASES
4      LCSEL ST      OPR1 OPR2 OPR3 STM1 STM2 STM3
5      AMOD
6      AMOD      STM1 1.333STM2 1.333STM3 1.333
7      * USE LOADING IN SEASTATE INPUT AND MODEL FILE
8      FILE B
9      CDM
10     CDM      1.0  0.600      1.200      0.600      1.200
11     CDM      100.0  0.600      1.200      0.600      1.200
12     MGROV
13     MGROV      0.00  200.000  1.000
14     MGROV      200.0  261.000  2.000
15     GRPOV
16     GRPOVAL      LG1 F
17     GRPOVAL      LG2 F
18     GRPOVAL      LG3 F
19     GRPOV      LG4 F
20     GRPOV      PL1NF      0.001 0.001
21     GRPOV      PL2NF      0.001 0.001
22     GRPOV      PL3NF      0.001 0.001
23     GRPOV      PL4NF      0.001 0.001
24     GRPOV      W.BNF 0.001      0.001 0.001
25     LOAD
26     LOADCNF000
27     WIND
28     WIND      50.00      0.0      AP13
29     WAVE
30     WAVE      STRE20.0      13.00      0.0      L  -75.  5.0  20MS      7
31     CURR
32     CURR      0.0  1.0      0.0      -15.0  BC NL FPS      AWP
33     CURR      261.0  2.0      0.0      -15.0  BC NL FPS      AWP
34     DEAD
35     DEAD      -Z      M
36     LOADCNF045
37     WIND
38     WIND      50.00      45.0      AP13
39     WAVE
40     WAVE      STRE20.0      13.00      45.0      L  -75.0 5.0  20MS      7
41     CURR
42     CURR      0.0  1.0      45.0      -15.0  BC NL FPS      AWP
43     CURR      261.0  2.0      45.0      -15.0  BC NL FPS      AWP
44     DEAD
45     DEAD      -Z      M
46     LOADCNF090
47     WIND
48     WIND      50.00      90.0      AP13
49     WAVE
50     WAVE      STRE20.0      13.00      90.0      L  -75.0 5.0  20MS      7
51     CURR
52     CURR      0.0  1.0      90.0      -15.0  BC NL FPS      AWP
53     CURR      261.0  2.0      90.0      -15.0  BC NL FPS      AWP
54     DEAD
55     DEAD      -Z      M
56     LOADCNS000
57     WIND
58     WIND      150.00      0.0 266.0  AP13
59     WAVE
```

```

59 WAVE STRE40.0 266.0 13.00 L-75. 5.0 20MS 7
60 CURR
61 CURR 0.0 1.0 0.0 -15.0 BC NL FPS AWP
62 CURR 261.0 3.5 0.0 -15.0 BC NL FPS AWP
63 DEAD
64 DEAD -Z 266.0 M
65 LOADCNS045
66 WIND
67 WIND 150.00 45.0 266.0 AP13
68 WAVE
69 WAVE STRE40.0 266.0 13.00 45.0 L -75.0 5.0 20MS 7
70 CURR
71 CURR 0.0 1.0 45.0 -15.0 BC NL FPS AWP
72 CURR 261.0 3.5 45.0 -15.0 BC NL FPS AWP
73 DEAD
74 DEAD -Z 266.0 M
75 LOADCNS090
76 WIND
77 WIND 150.00 90.0 266.0 AP13
78 WAVE
79 WAVE STRE40.0 266.0 13.00 90.0 L -75.0 5.0 20MS 7
80 CURR
81 CURR 0.0 1.0 90.0 -15.0 BC NL FPS AWP
82 CURR 261.0 3.5 90.0 -15.0 BC NL FPS AWP
83 DEAD
84 DEAD -Z 266.0 M
85 LCOMB
86 * OPERATIONAL COMBINATIONS
87 LCOMB OPR1 MISC 1.0 EQPT 1.0 AREA 0.5 LIVE 1.0 P000 1.0
88 LCOMB OPR2 MISC 1.0 EQPT 1.0 AREA 0.5 LIVE 1.0 P045 1.0
89 LCOMB OPR3 MISC 1.0 EQPT 1.0 AREA 0.5 LIVE 1.0 P090 1.0
90 * STORM COMBINATIONS
91 LCOMB STM1 MISC 1.0 EQPT 0.75 LIVE 0.75 S000 1.0
92 LCOMB STM2 MISC 1.0 EQPT 0.75 LIVE 0.75 S045 1.0
93 LCOMB STM3 MISC 1.0 EQPT 0.75 LIVE 0.75 S090 1.0
94 END

```

The results of the static analysis include a common solution file (sacsf.*) which contains the solution of the jacket stiffness matrix, however, the results for the deck elements are not included as the solution only contains the reduced stiffness matrix of the deck. The solution of the deck elements can be expanded since the original geometry of the deck is known. Performing a superelement expansion using the jacket common solution file and the two superelement expansion files generated by the deck superelement creation (sacpcs.* and psirst.*) will result in a new common solution file containing the solution of the deck elements.

The following are excerpts of the output listing file:

SACS-IV SYSTEM ELEMENT STRESS REPORT AT MAXIMUM UNITY CHECK																				
MEMBER	GRP	MAXIMUM UNITY CHECK	CRITICAL COND.	LOAD CASE NO.	DIST FROM END FT	***** APPLIED STRESSES *****					* CM VALUES *		* NEXT TWO HIGHEST CASES *							
						AXIAL	** BENDING **		*** SHEAR ***		Y	Z	Y	Z	Y	Z	UNITY CHECK	LOAD COND	UNITY CHECK	LOAD COND
							Y-Y KSI	Z-Z KSI	Y KSI	Z KSI										
601-	701 LG6	0.456	C<.15	STM1	0.00	-2.30	-0.87	12.62	0.88	-0.05	0.85	0.85	0.37	STM2	0.27	OPR1				
		0.539	C<.15	STM1	8.20	-3.23	-0.97	15.32	1.15	-0.09	0.85	0.85	0.44	STM2	0.33	OPR1				
		0.556	C<.15	STM1	8.20	-3.22	-0.96	15.26	1.15	-0.09	0.85	0.85	0.46	STM2	0.38	OPR3				
603-	703 LG6	0.520	C<.15	STM1	0.00	-3.27	-1.01	13.41	0.96	-0.02	0.85	0.85	0.44	STM2	0.31	OPR1				
		0.607	C>.15B	STM1	8.20	-4.60	-1.54	16.03	1.25	-0.04	0.85	0.85	0.51	STM2	0.38	OPR1				
		0.635	C>.15B	STM1	37.00	-4.51	0.05	-17.23	1.14	-0.04	0.85	0.85	0.54	STM2	0.42	STM3				
605-	705 LG6	0.401	C<.15	STM1	0.00	-1.42	0.62	12.01	0.82	0.02	0.85	0.85	0.35	STM2	0.31	STM3				
		0.483	C<.15	STM1	8.20	-2.00	0.47	14.86	1.06	0.05	0.85	0.85	0.41	STM2	0.34	STM3				
		0.493	C<.15	STM1	8.20	-2.00	0.47	14.81	1.06	0.05	0.85	0.85	0.45	STM3	0.43	STM2				
607-	707 LG6	0.564	C>.15A	STM1	0.00	-4.52	1.03	14.25	1.03	0.06	0.85	0.85	0.51	STM2	0.43	OPR1				
		0.692	C>.15B	STM1	8.20	-6.34	1.58	16.91	1.35	0.12	0.85	0.85	0.62	STM2	0.51	OPR1				
		0.748	C>.15B	STM1	37.00	-6.24	-0.01	-19.10	1.23	0.12	0.85	0.85	0.65	STM2	0.56	OPR1				
701-	801 LG7	1.126	C>.15B	OPR3	25.00	-3.55	0.35	25.96	1.50	-0.03	0.85	0.85	1.10	OPR2	1.09	OPR1				
703-	803 LG7	1.102	C>.15B	OPR1	25.00	-3.94	8.77	-23.24	1.63	0.02	0.85	0.85	1.09	OPR2	1.05	OPR3				
705-	805 LG7	0.779	C<.15	OPR3	25.00	-1.71	-1.82	18.61	1.18	-0.01	0.85	0.85	0.75	OPR2	0.73	OPR1				
707-	807 LG7	0.978	C>.15B	OPR1	25.00	-6.08	-9.40	-16.30	1.30	0.00	0.85	0.85	0.97	OPR2	0.95	OPR3				
701-	714 W01	0.606	TN+BN	OPR3	0.00	0.87	-11.56	0.41	-0.03	2.62	0.85	0.85	0.54	OPR2	0.53	STM2				
705-	717 W01	0.662	TN+BN	OPR3	0.00	0.63	-12.92	0.57	-0.03	2.96	0.85	0.85	0.59	STM3	0.58	OPR2				
714-	715 W01	0.506	TN+BN	OPR1	0.00	0.88	9.70	-0.06	0.00	-0.65	0.85	0.85	0.51	STM1	0.50	OPR2				
715-	703 W01	1.042	TN+BN	STM1	15.80	0.53	-28.98	-0.48	-0.02	-3.26	0.85	0.85	1.00	STM2	0.77	OPR1				
717-	718 W01	0.565	TN+BN	OPR1	0.00	0.62	11.16	-0.14	0.01	-0.64	0.85	0.85	0.56	OPR2	0.54	STM1				
718-	707 W01	1.054	TN+BN	STM1	15.80	0.31	-29.52	0.57	0.02	-3.49	0.85	0.85	0.97	STM2	0.78	OPR1				
SACS CONNECT Edition V(14.3) - CL																				
SAMPLE 13 ENGLISH UNITS MODEL SUPER ELEMENT (DECK)										Company: Bentley										
										DATE 23-OCT-2020 TIME 14:44:00 PST PAGE 6										
SACS-IV SYSTEM ELEMENT STRESS REPORT AT MAXIMUM UNITY CHECK																				
MEMBER	GRP	MAXIMUM UNITY CHECK	CRITICAL COND.	LOAD CASE NO.	DIST FROM END	***** APPLIED STRESSES *****					* CM VALUES *		* NEXT TWO HIGHEST CASES *							
						AXIAL	** BENDING **		*** SHEAR ***		Y	Z	Y	Z	UNITY CHECK	LOAD COND	UNITY CHECK	LOAD COND		
						Y-Y KSI	Z-Z KSI	Y KSI	Z KSI			Y	Z							

					FT	KSI	KSI	KSI	KSI	KSI							
801-	834	W01	1.363	C<.15	OPR1	15.80	-0.38	28.85	-0.10	-0.00	6.22	0.85	0.85	1.36	OPR2	1.35	OPR3
803-	836	W01	0.745	TN+BN	OPR3	0.00	0.00	-15.54	0.64	-0.02	1.86	0.85	0.85	0.74	OPR2	0.73	OPR1
805-	837	W01	0.785	C<.15	OPR3	0.00	-0.25	-16.42	-0.02	-0.01	4.06	0.85	0.85	0.78	OPR1	0.78	OPR2
807-	839	W01	1.210	C<.15	OPR3	0.00	-0.00	-25.17	0.79	-0.03	2.97	0.85	0.85	1.20	OPR2	1.18	OPR1
834-	835	W01	1.392	C<.15	OPR2	0.00	-0.40	28.77	0.21	-0.01	-1.24	0.85	0.85	1.39	OPR1	1.38	OPR3
835-	803	W01	1.754	C<.15	OPR1	15.80	-0.94	-36.05	-0.22	-0.01	-6.72	0.85	0.85	1.74	OPR2	1.72	OPR3
837-	838	W01	1.081	C<.15	OPR3	16.40	-0.25	22.30	-0.21	0.00	0.70	0.85	0.85	1.07	OPR2	1.06	OPR1
838-	807	W01	1.906	C<.15	OPR1	15.80	-0.73	-39.55	0.09	0.00	-7.68	0.85	0.85	1.90	OPR2	1.87	OPR3
701-	705	W02	0.904	BEND	STM3	32.00	0.00	-17.36	2.75	0.05	-3.07	0.85	0.85	0.71	STM2	0.57	OPR3
703-	707	W02	0.935	BEND	STM3	32.00	0.24	-16.99	-4.50	-0.07	-2.32	0.85	0.85	0.79	STM2	0.32	OPR3
705-	720	W02	0.123	TN+BN	STM1	0.00	0.00	-1.33	-2.92	0.15	0.53	0.85	0.85	0.10	STM2	0.08	OPR1
707-	723	W02	0.119	TN+BN	STM1	0.00	0.00	-1.21	-2.92	0.15	0.43	0.85	0.85	0.10	STM2	0.07	OPR1
709-	701	W02	0.108	TN+BN	STM1	10.25	0.00	-0.84	-2.92	-0.15	-0.30	0.85	0.85	0.08	STM2	0.06	OPR1
710-	714	W02	0.126	TN+BN	STM1	10.25	0.00	-1.43	-2.92	-0.15	-0.52	0.85	0.85	0.10	STM2	0.10	OPR1
711-	715	W02	0.126	TN+BN	STM1	10.25	0.00	-1.43	-2.92	-0.15	-0.52	0.85	0.85	0.10	STM2	0.10	OPR1
712-	703	W02	0.119	TN+BN	STM1	10.25	0.00	-1.21	-2.92	-0.15	-0.43	0.85	0.85	0.10	STM2	0.07	OPR1
714-	717	W02	0.919	BEND	OPR1	16.00	0.00	14.32	0.27	-0.00	-0.07	0.85	0.85	0.92	OPR2	0.91	OPR3
715-	718	W02	0.280	BEND	STM2	32.00	0.00	-1.92	-6.78	-0.23	-0.83	0.85	0.85	0.23	STM3	0.22	STM1
717-	721	W02	0.142	TN+BN	STM1	0.00	0.00	-1.92	-2.92	0.15	0.74	0.85	0.85	0.12	STM2	0.12	OPR1
718-	722	W02	0.142	TN+BN	STM1	0.00	0.00	-1.92	-2.92	0.15	0.74	0.85	0.85	0.12	STM2	0.12	OPR1
801-	805	W02	0.522	C<.15	OPR1	0.00	-0.04	-7.79	-0.52	0.03	3.44	0.85	0.85	0.49	OPR2	0.47	OPR3
803-	807	W02	1.143	C<.15	OPR3	32.00	-0.35	-17.00	-0.59	-0.01	-8.33	0.85	0.85	1.14	OPR2	1.09	OPR1

SACS-IV SYSTEM ELEMENT STRESS REPORT AT MAXIMUM UNITY CHECK

MEMBER	GRP	MAXIMUM	CRITICAL	LOAD	DIST	*****	APPLIED STRESSES				*****	* CM VALUES *		* NEXT TWO HIGHEST CASES *			
		UNITY CHECK	COND.	CASE NO.	FROM END FT	AXIAL KSI	** BENDING ** Y-Y KSI	Z-Z KSI	*** SHEAR *** Y KSI	Z KSI	Y	Z	UNITY CHECK	LOAD COND	UNITY CHECK	LOAD COND	
805-	840	W02	0.132	TN+BN	STM1	0.00	0.00	-1.43	-3.12	0.16	0.52	0.85	0.85	0.11	STM2	0.09	OPR1
807-	843	W02	0.256	TN+BN	OPR1	0.00	0.00	-5.78	-0.35	0.02	2.14	0.85	0.85	0.25	OPR2	0.24	OPR3
829-	801	W02	0.300	TN+BN	OPR1	10.25	0.00	-6.82	-0.35	-0.02	-3.11	0.85	0.85	0.30	OPR2	0.29	OPR3
830-	834	W02	0.393	TN+BN	OPR1	10.25	0.00	-9.02	-0.35	-0.02	-3.97	0.85	0.85	0.39	OPR2	0.38	OPR3
831-	835	W02	0.169	TN+BN	STM1	10.25	0.00	-2.61	-3.12	-0.16	-0.94	0.85	0.85	0.17	OPR1	0.16	OPR2
832-	803	W02	0.169	TN+BN	STM1	10.25	0.00	-2.61	-3.12	-0.16	-0.94	0.85	0.85	0.17	OPR1	0.16	OPR2
833-	836	W02	0.144	TN+BN	STM1	10.25	0.00	-1.80	-3.12	-0.16	-0.65	0.85	0.85	0.12	STM2	0.11	OPR1
834-	837	W02	0.593	BEND	OPR1	0.00	0.07	-9.00	-0.56	0.03	4.67	0.85	0.85	0.59	OPR2	0.57	OPR3
835-	838	W02	1.798	BEND	OPR1	16.00	0.03	28.16	0.29	0.00	2.66	0.85	0.85	1.80	OPR2	1.79	OPR3
836-	839	W02	0.294	BEND	OPR1	32.00	0.00	-4.20	-0.54	-0.03	-1.78	0.85	0.85	0.29	STM1	0.29	OPR2
837-	841	W02	0.169	TN+BN	STM1	0.00	0.00	-2.61	-3.12	0.16	0.94	0.85	0.85	0.17	OPR1	0.16	OPR2
838-	842	W02	0.169	TN+BN	STM1	0.00	0.00	-2.61	-3.12	0.16	0.94	0.85	0.85	0.17	OPR1	0.16	OPR2
839-	844	W02	0.191	TN+BN	OPR1	0.00	0.00	-4.22	-0.35	0.02	1.57	0.85	0.85	0.19	STM1	0.19	OPR2

4.3 DIRECT STIFFNESS INPUT

This example shows direct stiffness input. In this example, the 6X6 stiffness partitions for joints 1, 2, 3, 4, 101, 102, 103, and 104 including the off-diagonal partitions connecting joints 1 to 101, 2 to 102, 3 to 103, and 4 to 104 are input.

It should be noted that only one side of the stiffness matrix is input in that the partition 1 to 101 is input but not 101 to 1. The program requires that only one of the off-diagonal partitions is input and that partition is from the lowest joint name to the highest, i.e. joint 1 to 101.

The following is the Superelement input file followed by a brief description of input.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	SUBOPT INP MN							
2	STFHEAD 1 1	+1.0						
3	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.	
4	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.	
5	STFR FZ	0.	0.	1.045E6	0.	0.	0.	
6	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.	
7	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.	
8	STFR MZ	0.	0.	0.	0.	0.	5.656E3	
9	STFHEAD 2 2	+1.0						
10	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.	
11	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.	
12	STFR FZ	0.	0.	1.045E6	0.	0.	0.	
13	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.	
14	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.	
15	STFR MZ	0.	0.	0.	0.	0.	5.656E3	
16	STFHEAD 3 3	+1.0						
17	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.	
18	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.	
19	STFR FZ	0.	0.	1.045E6	0.	0.	0.	
20	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.	
21	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.	
22	STFR MZ	0.	0.	0.	0.	0.	5.656E3	
23	STFHEAD 4 4	+1.0						
24	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.	
25	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.	
26	STFR FZ	0.	0.	1.045E6	0.	0.	0.	
27	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.	
28	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.	
29	STFR MZ	0.	0.	0.	0.	0.	5.656E3	
30	STFHEAD 101 101	+1.0						
31	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.	
32	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.	
33	STFR FZ	0.	0.	1.045E6	0.	0.	0.	
34	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.	
35	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.	
36	STFR MZ	0.	0.	0.	0.	0.	5.656E3	
37	STFHEAD 102 102	+1.0						
38	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.	
39	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.	
40	STFR FZ	0.	0.	1.045E6	0.	0.	0.	
41	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.	
42	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.	
43	STFR MZ	0.	0.	0.	0.	0.	5.656E3	
44	STFHEAD 103 103	+1.0						
45	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.	
46	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.	
47	STFR FZ	0.	0.	1.045E6	0.	0.	0.	
48	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.	
49	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.	
50	STFR MZ	0.	0.	0.	0.	0.	5.656E3	
51	STFHEAD 104 104	+1.0						

52	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.
53	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.
54	STFR FZ	0.	0.	1.045E6	0.	0.	0.
55	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.
56	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.
57	STFR MZ	0.	0.	0.	0.	0.	5.656E3
58	STFHEAD	101	1	+1.0			
59	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.
60	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.
61	STFR FZ	0.	0.	1.045E6	0.	0.	0.
62	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.
63	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.
64	STFR MZ	0.	0.	0.	0.	0.	5.656E3
65	STFHEAD	102	2	+1.0			
66	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.
67	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.
68	STFR FZ	0.	0.	1.045E6	0.	0.	0.
69	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.
70	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.
71	STFR MZ	0.	0.	0.	0.	0.	5.656E3
72	STFHEAD	103	3	+1.0			
73	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.
74	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.
75	STFR FZ	0.	0.	1.045E6	0.	0.	0.
76	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.
77	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.
78	STFR MZ	0.	0.	0.	0.	0.	5.656E3
79	STFHEAD	104	4	+1.0			
80	STFR FX	.1342E6	0.	0.	0.	-9.4773E5	0.
81	STFR FY	0.	.13420E6	0.	9.4773E5	0.	0.
82	STFR FZ	0.	0.	1.045E6	0.	0.	0.
83	STFR MX	0.	9.4773E5	0.	0.9E7	0.	0.
84	STFR MY	-9.4773E5	0.	0.	0.	0.9E7	0.
85	STFR MZ	0.	0.	0.	0.	0.	5.656E3
86	END						

Line 1. The SUBOPT line indicates that a direct stiffness is input (ie. INP in columns 8-10).

Line 3. The STFHEAD lines identifies the partition using joint 1 as the first and second joint.

Line 3. The STFR line identifies the row as force in the X direction by FX in columns 6-7.

5 INPUT LINES

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE STIFFNESS MATRIX INPUT.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

STIFFNESS MATRIX PARTITION INPUT HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO INPUT STIFFNESS MATRICES DIRECTLY INTO SACS. IT SPECIFIES THE CONNECTION JOINTS FOR THE 6X6 STIFFNESS PARTITION WHICH WILL FOLLOW USING THE 'STFR**' LINES. FOR EXAMPLE, A DIAGONAL PARTITION WILL CONNECT A JOINT WITH ITSELF SO THE FIRST AND SECOND CONNECTING JOINTS WILL BE IDENTICAL. FOR OFF-DIAGONAL PARTITIONS, ONLY THE UPPER DIAGONAL PARTITIONS SHOULD BE ENTERED.
(10-13)	ENTER THE FIRST JOINT NAME TO IDENTIFY THIS PARTITION.
(14-17)	ENTER THE SECOND JOINT NAME TO IDENTIFY THIS PARTITION.
(18-27)	ENTER THE FACTOR TO BE APPLIED TO EACH TERM IN THE FOLLOWING STIFFNESS MATRIX. IF LEFT BLANK OR ZERO, A VALUE OF 1.0 WILL BE USED.

LINE LABEL	FIRST CONNECTING JOINT NAME	SECOND CONNECTING JOINT NAME	STIFFNESS FACTOR	LEAVE BLANK
STFHEAD				
1-- 7	10-->13	14-->17	18<--27	28-----80
DEFAULT			1	

STIFFNESS MATRIX PARTITION INPUT (EN UNITS)

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO INPUT STIFFNESS MATRICES DIRECTLY INTO SACS. EACH ROW OF THE 6X6 PARTITION REQUIRES A SEPARATE LINE AND ALL 6 ROWS MUST BE INPUT. THE ORDER OF THE INPUT OF THE ROWS IS IMMATERIAL.
(6- 7)	ENTER THE ROW IDENTIFICATION. USE 'F' FOR FORCE AND 'M' FOR MOMENT FOLLOWED BY THE DIRECTION. FOR EXAMPLE 'FX' IS THE ROW IDENTIFIER FOR THE FORCE IN THE X-DIRECTION.
(11-20)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT DISPLACEMENT (ONE INCH) IN THE X-DIRECTION.
(21-30)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT DISPLACEMENT (ONE INCH) IN THE Y-DIRECTION.
(31-40)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT DISPLACEMENT (ONE INCH) IN THE Z-DIRECTION.
(41-50)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT ROTATION (ONE RADIAN) IN THE X-DIRECTION.
(51-60)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT ROTATION (ONE RADIAN) IN THE Y-DIRECTION.
(61-70)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT ROTATION (ONE RADIAN) IN THE Z-DIRECTION.

LINE LABEL	ROW ID	STIFFNESS MATRIX TERMS					
		DEFLECTIONS			ROTATIONS		
		X-DIRECTION	Y-DIRECTION	Z-DIRECTION	X-DIRECTION	Y-DIRECTION	Z-DIRECTION
STFR							
1-- 4	6-- 7	11<--20	21<--30	31<--40	41<--50	51<--60	61<--70
DEFAULT							
FORCE		LB	LB	LB	LB	LB	LB
MOMENT		LB-IN	LB-IN	LB-IN	LB-IN	LB-IN	LB-IN

STIFFNESS MATRIX PARTITION INPUT (ME UNITS)

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO INPUT STIFFNESS MATRICES DIRECTLY INTO SACS. EACH ROW OF THE 6X6 PARTITION REQUIRES A SEPARATE LINE AND ALL 6 ROWS MUST BE INPUT. THE ORDER OF THE INPUT OF THE ROWS IS IMMATERIAL.
(6- 7)	ENTER THE ROW IDENTIFICATION. USE 'F' FOR FORCE AND 'M' FOR MOMENT FOLLOWED BY THE DIRECTION. FOR EXAMPLE 'FX' IS THE ROW IDENTIFIER FOR THE FORCE IN THE X-DIRECTION.
(11-20)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT DISPLACEMENT (ONE CM) IN THE X-DIRECTION.
(21-30)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT DISPLACEMENT (ONE CM) IN THE Y-DIRECTION.
(31-40)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT DISPLACEMENT (ONE CM) IN THE Z-DIRECTION.
(41-50)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT ROTATION (ONE RADIAN) IN THE X-DIRECTION.
(51-60)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT ROTATION (ONE RADIAN) IN THE Y-DIRECTION.
(61-70)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT ROTATION (ONE RADIAN) IN THE Z-DIRECTION.

LINE LABEL	ROW ID	STIFFNESS MATRIX TERMS					
		DEFLECTIONS			ROTATIONS		
		X-DIRECTION	Y-DIRECTION	Z-DIRECTION	X-DIRECTION	Y-DIRECTION	Z-DIRECTION
STFR							
1-- 4	6-- 7	11<--20	21<--30	31<--40	41<--50	51<--60	61<--70
DEFAULT							
FORCE		KG	KG	KG	KG	KG	KG
MOMENT		KG-CM	KG-CM	KG-CM	KG-CM	KG-CM	KG-CM

STIFFNESS MATRIX PARTITION INPUT (MN UNITS)

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO INPUT STIFFNESS MATRICES DIRECTLY INTO SACS. EACH ROW OF THE 6X6 PARTITION REQUIRES A SEPARATE LINE AND ALL 6 ROWS MUST BE INPUT. THE ORDER OF THE INPUT OF THE ROWS IS IMMATERIAL.
(6- 7)	ENTER THE ROW IDENTIFICATION. USE 'F' FOR FORCE AND 'M' FOR MOMENT FOLLOWED BY THE DIRECTION. FOR EXAMPLE 'FX' IS THE ROW IDENTIFIER FOR THE FORCE IN THE X-DIRECTION.
(11-20)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT DISPLACEMENT (ONE M) IN THE X-DIRECTION.
(21-30)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT DISPLACEMENT (ONE M) IN THE Y-DIRECTION.
(31-40)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT DISPLACEMENT (ONE M) IN THE Z-DIRECTION.
(41-50)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT ROTATION (ONE RADIAN) IN THE X-DIRECTION.
(51-60)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT ROTATION (ONE RADIAN) IN THE Y-DIRECTION.
(61-70)	ENTER THE FORCE OR MOMENT IN THE APPROPRIATE DIRECTION CAUSED BY A UNIT ROTATION (ONE RADIAN) IN THE Z-DIRECTION.

LINE LABEL	ROW ID	STIFFNESS MATRIX TERMS					
		DEFLECTIONS			ROTATIONS		
		X-DIRECTION	Y-DIRECTION	Z-DIRECTION	X-DIRECTION	Y-DIRECTION	Z-DIRECTION
STFR							
1-- 4	6-- 7	11<--20	21<--30	31<--40	41<--50	51<--60	61<--70
DEFAULT							
FORCE		KN	KN	KN	KN	KN	KN
MOMENT		KN-M	KN-M	KN-M	KN-M	KN-M	KN-M

SUBSTRUCTURE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO DIRECT THE PROGRAM AS TO WHICH FUNCTION TO PERFORM. THIS LINE MUST BE THE FIRST LINE IN THE DATA FILE.
(8-10)	SELECT THE DESIRED PRIMARY FUNCTION TO BE EXECUTED: 'RET' - RETRIEVE SUBSTRUCTURE FROM PSI FILE. 'XPN' - EXPAND SUBSTRUCTURE. 'MER' - MERGE TWO SUBSTRUCTURES. 'MOD' - MODIFY SUBSTRUCTURE. 'INP' - DIRECT STIFFNESS INPUT.
(12-13)	SELECT THE UNITS FROM THE FOLLOWING: 'EN' - ENGLISH UNITS (KIPS, INCHES, ETC.). 'MN' - METRIC UNITS WITH KILONEWTONS AS FORCE. 'ME' - METRIC UNITS WITH KILOGRAMS AS FORCE.

LINE LABEL	FUNCTION	UNITS	LEAVE BLANK
SUBOPT			
1-- 6	8--10	12--13	14-----80
DEFAULT		'EN'	

STIFFNESS MATRIX TRANSFORMATION DIRECT INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO INPUT STIFFNESS MATRIX TRANSFORMATIONS DIRECTLY TO BE USED WITH THE DIRECT INPUT OF THE STIFFNESS MATRIX PARTITIONS. THIS MATRIX WILL BE USED FOR ALL SUBSEQUENT STIFFNESS PARTITIONS UNTIL ANOTHER TRANSFORMATION MATRIX IS ENTERED. IF NO TRANSFORMATION MATRIX IS ENTERED, THEN NO TRANSFORMATION WILL BE PERFORMED. THE MATRIX OPERATION IS $K=TT^*K^*T$ WHERE TT IS THE TRANSPOSED TRANSFORMATION MATRIX, KO IS THE ORIGINAL STIFFNESS MATRIX, AND T IS THE TRANSFORMATION MATRIX.
(7)	ENTER 'D' TO SPECIFY THAT THIS IS A DIRECT INPUT OF THE TRANSFORMATION MATRIX.
(8-14)	ENTER THE TERM FOR THE FIRST ROW AND FIRST COLUMN IN THE TRANSFORMATION MATRIX.
(15-21)	TERM FOR FIRST ROW AND SECOND COLUMN.
(22-70)	ENTER REMAINING TERMS.

LINE LABEL	DIRECT INDICATOR	STIFFNESS MATRIX TRANSFORMATION TERMS								
		FIRST ROW			SECOND ROW			THIRD ROW		
		T11	T12	T13	T21	T22	T23	T31	T32	T33
TRAN										
1-- 4	7	8<--14	15<--21	22<--28	29<--35	36<--42	43<--49	50<--56	57<--63	64<--70

STIFFNESS MATRIX TRANSFORMATION VECTOR INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO INPUT STIFFNESS MATRIX TRANSFORMATIONS TO BE USED WITH THE DIRECT INPUT OF THE STIFFNESS MATRIX PARTITIONS. THIS MATRIX WILL BE USED FOR ALL SUBSEQUENT STIFFNESS PARTITIONS UNTIL ANOTHER TRANSFORMATION MATRIX IS ENTERED. IF NO TRANSFORMATION MATRIX IS ENTERED, THEN NO TRANSFORMATION WILL BE PERFORMED. THE MATRIX OPERATION IS $K=TT^*K^*T$ WHERE TT IS THE TRANSPOSED TRANSFORMATION MATRIX, KO IS THE ORIGINAL STIFFNESS MATRIX, AND T IS THE TRANSFORMATION MATRIX. THE MATRIX IS DEFINED BY THE INPUT POINTS SUCH THAT THE X-DIRECTION IS DEFINED AS THE VECTOR FROM POINT 1 TO POINT 2. THE Y-DIRECTION IS DETERMINED BY A VECTOR CREATED BY TAKING THE CROSS PRODUCT OF THE VECTOR FROM POINT 1 TO POINT 3 INTO THE VECTOR FROM POINT 1 TO POINT 2. THE Z-DIRECTION IS THEN FORMED BY THE RIGHT-HAND RULE.
(7)	ENTER 'I' TO SPECIFY THAT THE TRANSFORMATION MATRIX IS TO BE INVERTED (TRANSPOSED) BEFORE USING. OTHERWISE, LEAVE BLANK.
(8-70)	ENTER THE COORDINATES OF THE POINTS. THE UNITS ARE IMMATERIAL SINCE DIRECTIONS ARE ALL THAT IS BEING DEFINED.

LINE LABEL	INVERSE INDICATOR	TRANSFORMATION VECTOR COORDINATES								
		FIRST POINT			SECOND POINT			THIRD POINT		
		X1	Y1	Z1	X2	Y2	Z2	X3	Y3	Z3
TRAN										
1-- 4	7	8<--14	15<--21	22<--28	29<--35	36<--42	43<--49	50<--56	57<--63	64<--70

DEGREES OF FREEDOM ORDER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL TO CHANGE ORDER OF DEGREES OF FREEDOMS IN STIFFNESS MATRIX AND FORCE VECTOR.
(8-13)	ENTER A SIX-DIGIT NUMBER TO CHANGE THE ORDER OF OUTPUT DEGREES OF FREEDOM.

LINE LABEL	OUTPUT DOF ORDER	LEAVE BLANK
DOFORD		
1-- 6	8--13	14-----80
DEFAULT		

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE STIFFNESS MATRIX INPUT.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

SUBSTRUCTURE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO DIRECT THE PROGRAM AS TO WHICH FUNCTION TO PERFORM. THIS LINE MUST BE THE FIRST LINE IN THE DATA FILE.
(8-10)	SELECT THE DESIRED PRIMARY FUNCTION TO BE EXECUTED: 'RET' - RETRIEVE SUBSTRUCTURE FROM PSI FILE. 'XPN' - EXPAND SUBSTRUCTURE. 'MER' - MERGE TWO SUBSTRUCTURES. 'MOD' - MODIFY SUBSTRUCTURE. 'INP' - DIRECT STIFFNESS INPUT. 'FRM' - CONVERT SACS SUPERELEMENT INTO A FORMATTED TEXT FILE. 'UNF' - CONVERT A FORMATTED TEXT FILE INTO SACS SUPERELEMENT.
(12-13)	SELECT THE UNITS FROM THE FOLLOWING: 'EN' - ENGLISH UNITS (KIPS, INCHES, ETC.). 'MN' - METRIC UNITS WITH KILONEWTONS AS FORCE. 'ME' - METRIC UNITS WITH KILOGRAMS AS FORCE.
(17-18)	ENTER THE DIRECTION OF VERTICAL COORDINATES (+X,-X,+Y,-Y,+Z,-Z) FOR SUPERELEMENT.

LINE LABEL	FUNCTION	UNITS	VERTICAL COODINATES DIRECTION	LEAVE BLANK
SUBOPT				
1-- 6	8--10	12--13	17--18	19-----80
DEFAULT		'EN'	'+Z'	

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE STIFFNESS MATRIX INPUT.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

MERGE LOADS

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO MERGE THE LOAD CASES FROM TWO SUPERELEMENT FILES. THIS LINE CAN BE REPEATED AS MANY TIMES AS NECESSARY TO MERGE ALL REQUIRED LOAD CASES.
(9-12)	ENTER THE LOAD CASE NAME FROM THE FIRST SUBSTRUCTURE TO MERGE WITH THE LOAD CASE FROM THE SECOND SUBSTRUCTURE. LEAVE BLANK IF THE LOAD FROM THE SECOND SUBSTRUCTURE WILL BE ONLY MERGED TO THE OUTPUT SUBSTRUCTURE.
(13-16)	ENTER THE LOAD CASE NAME FROM THE SECOND SUBSTRUCTURE TO MERGE WITH THE LOAD CASE FROM THE FDIRST SUBSTRUCTURE. . LEAVE BLANK IF THE LOAD FROM THE FIRST SUBSTRUCTURE WILL BE ONLY MERGED TO THE OUTPUT SUBSTRUCTURE.
(17-80)	ENTER THE REMAINING PAIRS OF LOAD CASES FROM FIRST ABD SECOND SUBSTRUCTURE FILES. NOTE: THE LOAD CASES IN THE OUTPUT SUBSTRUCTURE WILL BE THE SAME AS THE LOAD NAME IN THE FIRST SUBSTRUCTURE, IF IT IS NOT BLANK. OTHERWISE, THE OUTPUT LOAD CASE NAME IS THE SAME AS THE SECOND SUBSTRUCTURE. NOTE: THE PROGRAM WILL IGNORE ANY DATA BEYOND THE FIRST PAIR OF BLANK LOAD CASE NAMES ENCOUNTERED ON THIS LINE. NOTE: THE PROGRAM ONLY INCLUDES THE LOAD CASES ENTERED IN THE LINE. OTHER LOAD CASES ARE NOT TRANSFERRED TO THE NEW SUBSTRUCTURE. THEREFORE, ALL LOAD CASES THAT ARE TO BE INCLUDED IN THE NEW SUBSTRUCTURE MUST BE SPECIFIED.

LINE LABEL	1ST LOAD CASE PAIR		2ND LOAD CASE PAIR		3RD LOAD CASE PAIR		4TH LOAD CASE PAIR		5TH LOAD CASE PAIR		6TH LOAD CASE PAIR		7TH LOAD CASE PAIR		8TH LOAD CASE PAIR		9TH LOAD CASE PAIR	
	1ST FILE	2ND FILE	1ST FILE	2ND FILE	1ST FILE	2ND FILE	1ST FILE	2ND FILE	1ST FILE	2ND FILE	1ST FILE	2ND FILE	1ST FILE	2ND FILE	1ST FILE	2ND FILE	1ST FILE	2ND FILE
MRGLOAD																		
1-- 7	9--12	13--16	17--20	21--24	25--28	29--32	33--36	37--40	41--44	45--48	49--52	53--56	57--60	61--64	65--68	69--72	73--76	77--80

SUBSTRUCTURE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO DIRECT THE PROGRAM AS TO WHICH FUNCTION TO PERFORM. THIS LINE MUST BE THE FIRST LINE IN THE DATA FILE.
(8-10)	SELECT THE DESIRED PRIMARY FUNCTION TO BE EXECUTED: 'RET' - RETRIEVE SUBSTRUCTURE FROM PSI FILE. 'XPN' - EXPAND SUBSTRUCTURE. 'MER' - MERGE TWO SUBSTRUCTURES. 'MOD' - MODIFY SUBSTRUCTURE. 'INP' - DIRECT STIFFNESS INPUT.
(12-13)	SELECT THE UNITS FROM THE FOLLOWING: 'EN' - ENGLISH UNITS (KIPS, INCHES, ETC.). 'MN' - METRIC UNITS WITH KILONEWTONS AS FORCE. 'ME' - METRIC UNITS WITH KILOGRAMS AS FORCE.

LINE LABEL	FUNCTION	UNITS	LEAVE BLANK
SUBOPT			
1-- 6	8--10	12--13	14-----80
DEFAULT		'EN'	

CHANGE JOINT NAMES

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO CHANGE THE JOINT NUMBERING FROM THE OLD SUBSTRUCTURE TO THE NEW. IT CAN BE USED IN CONJUNCTION WITH THE 'CHGL' LINES TO MODIFY LOAD CASES AND RENAME JOINTS TO CREATE A NEW SUBSTRUCTURE FROM AN EXISTING ONE. THIS LINE CAN BE REPEATED AS MANY TIMES AS NECESSARY TO MODIFY THE REQUIRED JOINTS.
(9-12)	ENTER THE JOINT NAME FROM THE ORIGINAL SUBSTRUCTURE.
(13-16)	ENTER THE JOINT NAME FOR THE NEW SUBSTRUCTURE.
(17-80)	ENTER THE REMAINING PAIRS OF ORIGINAL AND NEW JOINT NAMES. NOTE: THE PROGRAM WILL IGNORE ANY DATA BEYOND THE FIRST PAIR OF BLANK JOINT NAMES ENCOUNTERED ON THIS LINE. NOTE: FOR THE JOINTS ARE NOT LISTED ON THIS LINE, THE PROGRAM USES THEIR NAMES IN THE ORIGINAL SUBSTRUCUTRE.

LINE LABEL	1ST JOINT		2ND JOINT		3RD JOINT		4TH JOINT		5TH JOINT		6TH JOINT		7TH JOINT		8TH JOINT		9TH JOINT	
	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME
CHGJ																		
1-- 4	9--12	13--16	17--20	21--24	25--28	29--32	33--36	37--40	41--44	45--48	49--52	53--56	57--60	61--64	65--68	69--72	73--76	77--80

CHANGE LOAD CASE NAMES

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO CHANGE THE LOAD CASE NAMING FROM THE OLD SUBSTRUCTURE TO THE NEW. IT CAN BE USED IN CONJUNCTION WITH THE 'CHGJ' LINES TO MODIFY LOAD CASES AND RENAME JOINTS TO CREATE A NEW SUBSTRUCTURE FROM AN EXISTING ONE. THIS LINE CAN BE REPEATED AS MANY TIMES AS NECESSARY TO MODIFY THE REQUIRED LOAD CASES.
(9-12)	ENTER THE LOAD CASE NAME FROM THE ORIGINAL SUBSTRUCTURE OR LEAVE BLANK TO ADD A ZERO (EMPTY)LOAD CASE TO THE NEW SUBSTRUCTURE.
(13-16)	ENTER THE LOAD CASE NAME FOR THE NEW SUBSTRUCTURE.
(17-80)	ENTER THE REMAINING PAIRS OF ORIGINAL AND NEW LOAD CASE NAMES. NOTE: THE PROGRAM WILL IGNORE ANY DATA BEYOND THE FIRST PAIR OF BLANK LOAD CASE NAMES ENCOUNTERED ON THIS LINE. NOTE: THE PROGRAM ONLY INCLUDES THE LOAD CASES ENTERED IN THE LINE. OTHER LOAD CASES ARE NOT TRANSFERED TO THE NEW SUBSTRUCTURE. THEREFORE, ALL LOAD CASES THAT ARE TO BE INCLUDED IN THE NEW SUBSTRUCTURE MUST BE SPECIFIED EVEN THOUGH ITS LOAD CASE NAME IS THE SAME.

LINE LABEL	1ST LOAD CASE		2ND LOAD CASE		3RD LOAD CASE		4TH LOAD CASE		5TH LOAD CASE		6TH LOAD CASE		7TH LOAD CASE		8TH LOAD CASE		9TH LOAD CASE	
	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME	ORIGINAL NAME	NEW NAME
CHGL																		
1-- 4	9--12	13--16	17--20	21--24	25--28	29--32	33--36	37--40	41--44	45--48	49--52	53--56	57--60	61--64	65--68	69--72	73--76	77--80

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE STIFFNESS MATRIX INPUT.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

SUBSTRUCTURE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO DIRECT THE PROGRAM AS TO WHICH FUNCTION TO PERFORM. THIS LINE MUST BE THE FIRST LINE IN THE DATA FILE.
(8-10)	SELECT THE DESIRED PRIMARY FUNCTION TO BE EXECUTED: 'RET' - RETRIEVE SUBSTRUCTURE FROM PSI FILE. 'XPN' - EXPAND SUBSTRUCTURE. 'MER' - MERGE TWO SUBSTRUCTURES. 'MOD' - MODIFY SUBSTRUCTURE. 'INP' - DIRECT STIFFNESS INPUT.
(12-13)	SELECT THE UNITS FROM THE FOLLOWING: 'EN' - ENGLISH UNITS (KIPS, INCHES, ETC.). 'MN' - METRIC UNITS WITH KILONEWTONS AS FORCE. 'ME' - METRIC UNITS WITH KILOGRAMS AS FORCE.

LINE LABEL	FUNCTION	UNITS	LEAVE BLANK
SUBOPT			
1-- 6	8--10	12--13	14-----80
DEFAULT		'EN'	

DEGREES OF FREEDOM ORDER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL TO CHANGE ORDER OF DEGREES OF FREEDOMS IN STIFFNESS MATRIX AND FORCE VECTOR.
(8-13)	ENTER A SIX-DIGIT NUMBER TO CHANGE THE ORDER OF OUTPUT DEGREES OF FREEDOM.

LINE LABEL	OUTPUT DOF ORDER	LEAVE BLANK
DOFORD		
1-- 6	8--13	14-----80
DEFAULT		

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE STIFFNESS MATRIX INPUT.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

INTERFACE JOINTS HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO ENTER JOINT IDS OF SUPERELEMENT.
(8-10)	ENTER NUMBER OF INTERFACE JOINTS IN SUPERELEMENT.

LINE LABEL	NUMBER OF INTERFACE JOINTS	LEAVE BLANK
INTJNT		
1-- 6	8--10	11-----80
DEFAULT		

INTERFACE JOINT ID

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO ENTER INTERFACE JOINTS ID. THIS INPUT LINE NEEDS INTJNT HEADER LINE.
(8-11)	ENTER THE INTERFACE JOINT ID IN SUPERELEMENT.

LINE LABEL	INPUT INTERFACE JOINT ID	LEAVE BLANK
JOINT		
1-- 6	8--11	12-----80
DEFAULT		

LOAD CONDITIONS HEADER

COLUMNS COMMENTARY

GENERAL THIS LINE IS REQUIRED TO ENTER LOAD CONDITIONS OF SUPERELEMENT.

(8-10) ENTER NUMBER OF LOAD CONDITIONS IN SUPERELEMENT.

LINE LABEL	NUMBER OF LOAD CONDITIONS	LEAVE BLANK
LOAD		
1-- 6	8--10	11-----80
DEFAULT		

LOAD CONDITION NAME

COLUMNS COMMENTARY

GENERAL THIS LINE IS REQUIRED TO ENTER LOAD CONDITION NAME.
 THIS INPUT LINE NEEDS LOAD HEADER LINE.

(8-11) ENTER THE LOAD CONDITION NAME IN SUPERELEMENT.

LINE LABEL	INPUT LOAD CONDITION NAME	LEAVE BLANK
LOADCN		
1-- 6	8--11	12-----80
DEFAULT		

SUBSTRUCTURE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO DIRECT THE PROGRAM AS TO WHICH FUNCTION TO PERFORM. THIS LINE MUST BE THE FIRST LINE IN THE DATA FILE.
(8-10)	SELECT THE DESIRED PRIMARY FUNCTION TO BE EXECUTED: 'RET' - RETRIEVE SUBSTRUCTURE FROM PSI FILE. 'XPN' - EXPAND SUBSTRUCTURE. 'MER' - MERGE TWO SUBSTRUCTURES. 'MOD' - MODIFY SUBSTRUCTURE. 'INP' - DIRECT STIFFNESS INPUT. 'FRM' - CONVERT SACS SUPERELEMENT INTO A FORMATTED TEXT FILE. 'UNF' - CONVERT A FORMATTED TEXT FILE INTO SACS SUPERELEMENT.
(12-13)	SELECT THE UNITS FROM THE FOLLOWING: 'EN' - ENGLISH UNITS (KIPS, INCHES, ETC.). 'MN' - METRIC UNITS WITH KILONEWTONS AS FORCE. 'ME' - METRIC UNITS WITH KILOGRAMS AS FORCE.
(17-18)	ENTER THE DIRECTION OF VERTICAL COORDINATES (+X,-X,+Y,-Y,+Z,-Z) FOR SUPERELEMENT.

LINE LABEL	FUNCTION	UNITS	VERTICAL COODINATES DIRECTION	LEAVE BLANK
SUBOPT				
1-- 6	8--10	12--13	17--18	19-----80
DEFAULT		'EN'	'+Z'	